Solutions for exercises after class 8

1. The confidence intervals based on the data for 90%, 95%, 98% confidence levels respectively:

$$81.2 \pm 1.328 \cdot \frac{\sqrt{26.5}}{\sqrt{20}}, \quad 81.2 \pm 1.990 \cdot \frac{\sqrt{26.5}}{\sqrt{20}}$$

$$81.2 \pm 2.522 \cdot \frac{\sqrt{26.5}}{\sqrt{20}}.$$

2. Below is an advanced approach to the problem. We can apply the chi-squared test in this case. The p-value is the least significance α at which the null hypotheses is still rejected. The sketch of the theory:

1 Chi-squared test of homogeneity

Suppose we have an m by n array of random variables X_{ij} such that the rows are independent and they follow multinomial distributions (therefore the sum of their components are fixed, but possibly different) with probability vector $p_i = (p_{i1}, \dots, p_{in})$. We want to test that the rows are equally distributed, i.e. we have

$$H_0 : \ p_1 = \cdots = p_m, \sum_j p_{ij} = 1 \qquad H_1 : \ \sum_j p_{ij} = 1$$

Applying the Lagrangian technique as in the goodness-of-fit case we can find the MLEs for p_{ij} under H_0 as $\hat{p}_{ij} = \frac{x_{\cdot j}}{x_{\cdot\cdot}}$ and under H_1 as $\hat{\hat{p}}_{ij} = \frac{x_{ij}}{x_{i\cdot}}$, where $x_{i\cdot} = \sum_{t=1}^{n} x_{it}$, $x_{\cdot j} = \sum_{s=1}^{n} x_{sj}$ and $x_{\cdot\cdot} = \sum_{s=1}^{m} \sum_{t=1}^{n} x_{st}$. Now if $o_{ij} = x_{ij}$ and

$$e_{ij} = \hat{p}_{ij} x_{i\cdot} = \frac{x_{i\cdot} x_{\cdot j}}{x_{\cdot\cdot}}$$

then we have $-2 \log \frac{\sup_{H_0} f(x|p)}{\sup_{H_1} f(x|p)} = -2 \sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} \log \frac{\hat{p}_{ij}}{\hat{\hat{p}}_{ij}} = 2 \sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} \log \frac{x_{ij} x_{\cdot\cdot}}{x_{i\cdot} x_{\cdot j}} = 2 \sum_{i=1}^{m} \sum_{j=1}^{n} o_{ij} \log \frac{o_{ij}}{e_{ij}}$ which is approximately χ^2 - distributed with $(m-1)(n-1)$ degrees of freedom since we had $(m-1)(n-1)$ more independent restrictions on the parameters in H_0 than we did in H_1. Similarly to the goodness-of-fit case, we have that

$$\sum_{i=1}^{m} \sum_{j=1}^{n} \frac{(o_{ij} - e_{ij})^2}{e_{ij}} \tag{1}$$

is approximately chi-squared distributed with $(m-1)(n-1)$ degrees of freedom.

And its application:

	Heart attack	No heart attack	Total
Aspirin	104	10,933	11,037
Placebo	189	10,845	11,034
Total	293	21,778	22,071

Figure 1: The experiment on heart attack

	Heart attack	No heart attack	Total
Aspirin	104(146.52)	10,933(10890.5)	11,037
Placebo	189(146.48)	10,845(10887.5)	11,034
Total	293	21,778	22,071

Figure 2: The completion of Table 1 with the expected counts.

For example, $e_{11} = \frac{11{,}037 \cdot 293}{22{,}071} = 146.52$. The χ^2 statistic:

$$\frac{(104-146.52)^2}{146.52} + \frac{(189-146.48)^2}{146.48} + \frac{(10933-10890.5)^2}{10890.5} + \frac{(10845-10887.5)^2}{10887.5} = 25.01.$$

The degree of freedom for the chi-squared distribution is $(2-1)(2-1) = 1$ and the p -value is $5.69 \cdot 10^{-7}$. Therefore we reject the null hypotheses that the rate of heart attack is independent of whether the subject did or did not take the aspirin.